
Lab 2

Data Definition and Data Manipulation

CSE 4308
DATABASE MANAGEMENT SYSTEMS LAB

AUGUST 16, 2023

1 Data Definition

1.1 Creating a Table

The general syntax for creating a table is as follows:

```
CREATE TABLE table_name
(
attribute1 datatype [ NULL | NOT NULL ],
attribute2 datatype [ NULL | NOT NULL ],
...
);
```

There exist different data types in Oracle. Some of them are as follows:

- `char(n)`: value contains exactly n alpha-numeric characters
- `varchar2(n)`: value contains at most n alpha-numeric characters
- `number`: any integer or real numbers
- `date`: DD-MON-YY format, like '20-JAN-22'

Assume that you want to create a table named 'TEST' with 3 attributes:

1. `national_id`: number type
2. `name`: `varchar2` type
3. `birth_date`: date type

To create this table, you have to write the following query:

```
CREATE TABLE CITIZEN
(
  NATIONAL_ID NUMBER NOT NULL,
  NAME VARCHAR2(50) NOT NULL,
  BIRTH_DATE DATE
);
```

To create a table with constraints, the syntax is given below:

```
CREATE TABLE table_name
(
attribute1 datatype [ NULL | NOT NULL ],
attribute2 datatype [ NULL | NOT NULL ],
...,
[CONSTRAINT constraint_name] PRIMARY KEY (primary_attribute1, ...),
[CONSTRAINT constraint_name] CHECK condition
);
```

Primary key is a special column that is able to uniquely identify each record. For example, if we want to create a table called CITIZEN, it might have the NATIONAL_ID as the primary key.

```
CREATE TABLE CITIZEN
(
  NATIONAL_ID NUMBER NOT NULL ,
  NAME VARCHAR2(50) NOT NULL ,
  AGE INT ,
  COUNTRY VARCHAR2(20) ,
  CONSTRAINT PK_COURSE PRIMARY KEY(NATIONAL_ID) ,
  CONSTRAINT VALIDITY_CHECK CHECK (AGE > 17 AND COUNTRY= '
  BANGLADESH')
);
```

1.2 Dropping Tables

To delete the schema, we use:

```
DROP TABLE table_name;
```

It will also delete the information stored in the table.

To delete the table structure with constraints, we generally follow this command:

```
DROP TABLE table_name CASCADE CONSTRAINTS;
```

```
DROP TABLE CITIZEN CASCADE CONSTRAINTS ;
```

1.3 Altering Tables

To add a new attribute to the table, we use:

```
ALTER TABLE table_name ADD attribute_name datatype;
```

We can even add multiple attributes at the same time:

```
ALTER TABLE table_name ADD (attribute1 datatype, ...);
```

```
ALTER TABLE CITIZEN ADD PHONE_NO VARCHAR2(12);
```

To delete an attribute from a table, we use:

```
ALTER TABLE table_name DROP COLUMN attribute_name;
```

We can even delete multiple attributes at the same time:

```
ALTER TABLE table_name DROP COLUMN (attribute1, ...);
```

```
ALTER TABLE CITIZEN DROP COLUMN PHONE_NO ;
```

To modify the data type of an attribute, we need to ensure that the column is empty. Then we can execute:

```
ALTER TABLE table_name MODIFY attribute_name new_datatype;
```

```
ALTER TABLE CITIZEN MODIFY AGE NUMBER(2, 1);
```

To rename an attribute, we use:

```
ALTER TABLE table_name RENAME COLUMN old_attribute_name TO new_attribute_name;
```

To rename a table, we use:

```
ALTER TABLE table_name RENAME TO new_table_name;
```

2 Data Manipulation

2.1 Inserting Records/Rows into a Table

The general format for inserting a new record is:

```
INSERT INTO TABLE_NAME VALUES (... , ... , ...);
```

We can type the following command to insert new records into our 'TEST' table using positional notation:

```
INSERT INTO CITIZEN VALUES (2015001, 'W', 19, 'Bangladesh');
```

We can insert new records into 'TEST' table following named notation:

```
INSERT INTO CITIZEN (NATIONAL_ID, NAME, AGE, COUNTRY) VALUES  
(2015002, 'X', 23, 'Bangladesh');
```

2.2 Retrieval of Information

The basic SQL syntax of a query is as follows:

```
SELECT ATTRIBUTE1, ATTRIBUTE2  
FROM TABLE_NAME  
WHERE <CONDITIONAL CLAUSE>;
```

For example, to see the 'national_id' of all citizens from the 'CITIZEN' table:

```
SELECT NATIONAL_ID FROM CITIZEN;
```

To find the information of a citizen following a condition:

```
SELECT NATIONAL_ID FROM CITIZEN WHERE ID = 2015001;
```

To find the information of a citizen following multiple conditions:

```
SELECT NATIONAL_ID FROM CITIZEN WHERE AGE > 21 AND COUNTRY = '  
Bangladesh';
```

```
SELECT NATIONAL_ID FROM CITIZEN WHERE AGE > 21 OR NAME = 'W';
```

To find all records and their information:

```
SELECT * FROM CITIZEN;
```

3 Executing SQL Script in SQLPlus

You can create a file with .sql extension that contains your sql statements. Then you can execute it from the SQLPlus command line directly.

Suppose, you have written your SQL statements in a file 'a.sql' saved under 'd:\sample' directory. To execute that script, you have to type:

```
@d:\sample\a.sql
```

4 Lab Task

You have to write all SQL statements in an editor first and save them with .sql extension. Then execute the SQL script.

1. Create a user with username = <student_id> and password = cse4308 and grant necessary privileges to log in and execute DDL and DML statements. Then log in as that user.

2. Write SQL statement to create a table 'STUDENT' which has 4 attributes:

- ID (assign it as Primary Key)
- NAME
- DEPT_NAME
- TOT_CRED (ensure that TOT_CRED is not greater than 180)

3. Write SQL statements to insert the following records into 'STUDENT' table:

ID	NAME	DEPT_NAME	TOT_CRED
00128	Zhang	Comp. Sci.	102
12345	Shankar	Comp. Sci.	32
19991	Brandt	History	80
23121	Chavez	Finance	110
44553	Peltier	Physics	56
45678	Levy	Physics	46
54321	Williams	Comp. Sci.	5
55739	Sanchez	Music	38
70557	Snow	Physics	0
76543	Brown	Comp. Sci.	58
76653	Aoi	Elec. Eng.	60
98765	Bourikas	Elec. Eng.	9
98988	Tanaka	Biology	120

4. Write SQL statements to perform the following queries:

- (a) Display all records of 'STUDENT' table.
- (b) Show student ID and name only.
- (c) Find name and department of students who have completed more than 100 credits.
- (d) Find name and department of students who have completed in between 80 and 120 credits (inclusive).
- (e) Find ID and name of students of Comp. Sci. department.
- (f) Find name and total credit of students of Physics department.
- (g) Find ID and name of students of Comp. Sci. department or students who have completed less than 10 credits.
- (h) Find the names of the department.

5. Drop the 'Student' table with all its constraints.